2-4 BOX PLOT

- While tools like histograms show general trends and numerical values focus on specific stats, a **box plot** offers a multi-dimensional view of a data set in a single graphic.

  - **Comprehensive Summary**: A **box plot** simultaneously highlights the center (median) and the spread of the data.

  - **Symmetry Assessment**: It allows users to quickly see if the data is symmetric or skewed based on the position of the box and whiskers.

- **Outlier Detection**: The display is specifically designed for the identification of observations that lie unusually far from the rest of the data.

- A **box plot**, also known as **box-and-whisker plots**, provides a visual summary of data using quartiles and specific markers for unusual values, as illustrated in **Fig. 2-12**.

  - **Box Construction**: A rectangular box represents the interquartile range (IQR), with edges at the first (q_1) and third (q_3) quartiles and a line marking the median.

  - **Whisker Limits**: A **whisker** extends from each side of the box to the smallest and largest data points located within 1.5 times the IQR.

- **Outlier Identification**: Any data point falling beyond the whiskers but within 3 IQRs of the box edge is classified as an **outlier**.

- **Extreme Values**: A point falling more than 3 IQRs away from the box edge is labeled an **extreme outlier**.

- **Analytical Value**: In **box-and-whisker plots**, these atypical points deserve investigation as they may represent recording errors or significant, unique findings.

- **Figure 2-13** displays a Minitab-generated box plot for alloy compressive strength, offering a visual breakdown of its distribution.

  - **Symmetry**: The plot shown in **Figure 2-13** suggests the data is fairly symmetric, as the median sits centrally and the whiskers are roughly equal in length.

  - **Box Proportions**: The lengths of the boxes on either side of the median are balanced, further supporting the observation of a symmetric distribution.

- **Atypical Values**: The analysis of **Figure 2-13** reveals the presence of outliers at both the high and low ends of the data range.

Figure 2-13: Compressive Strength Boxplot (Base R)

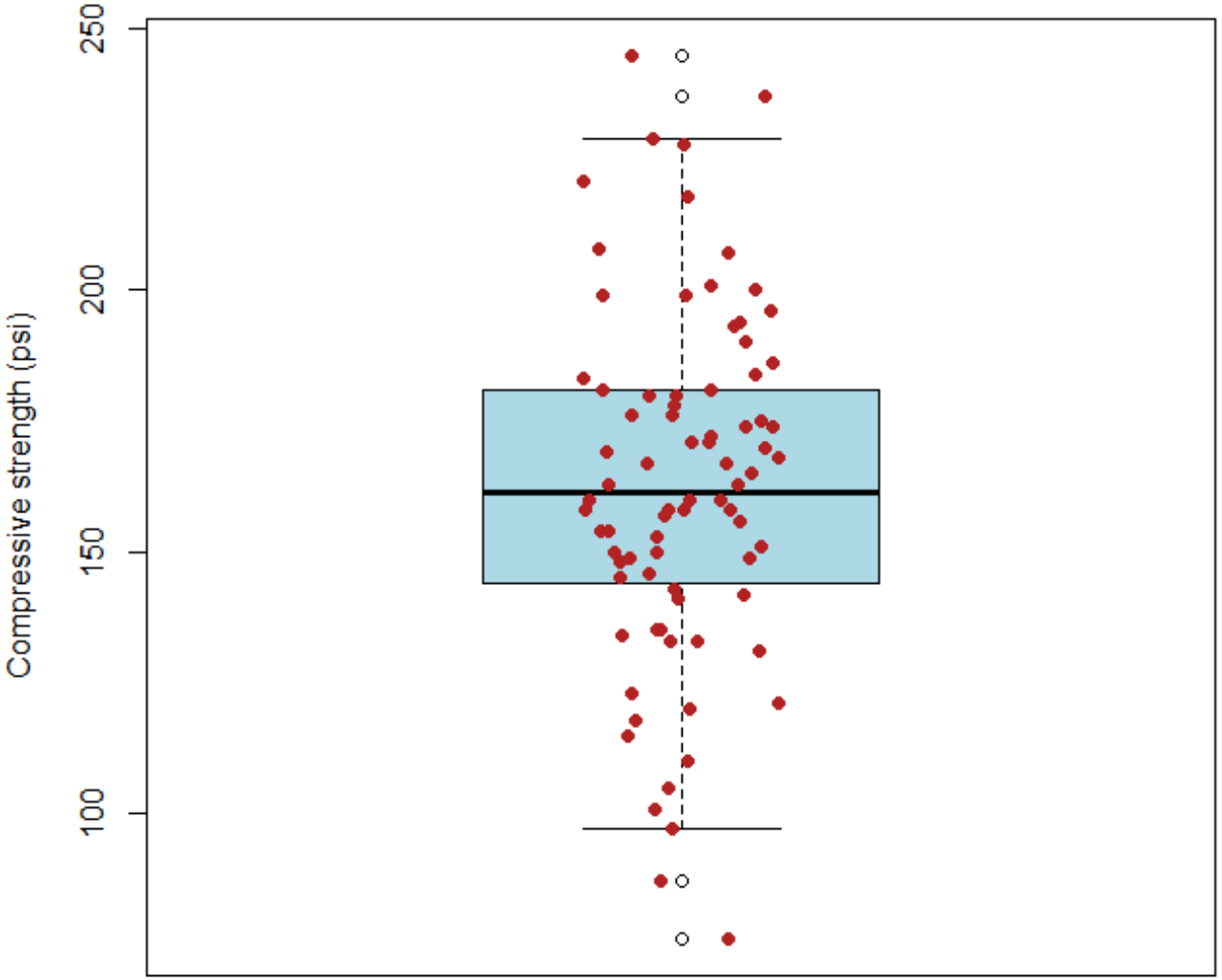
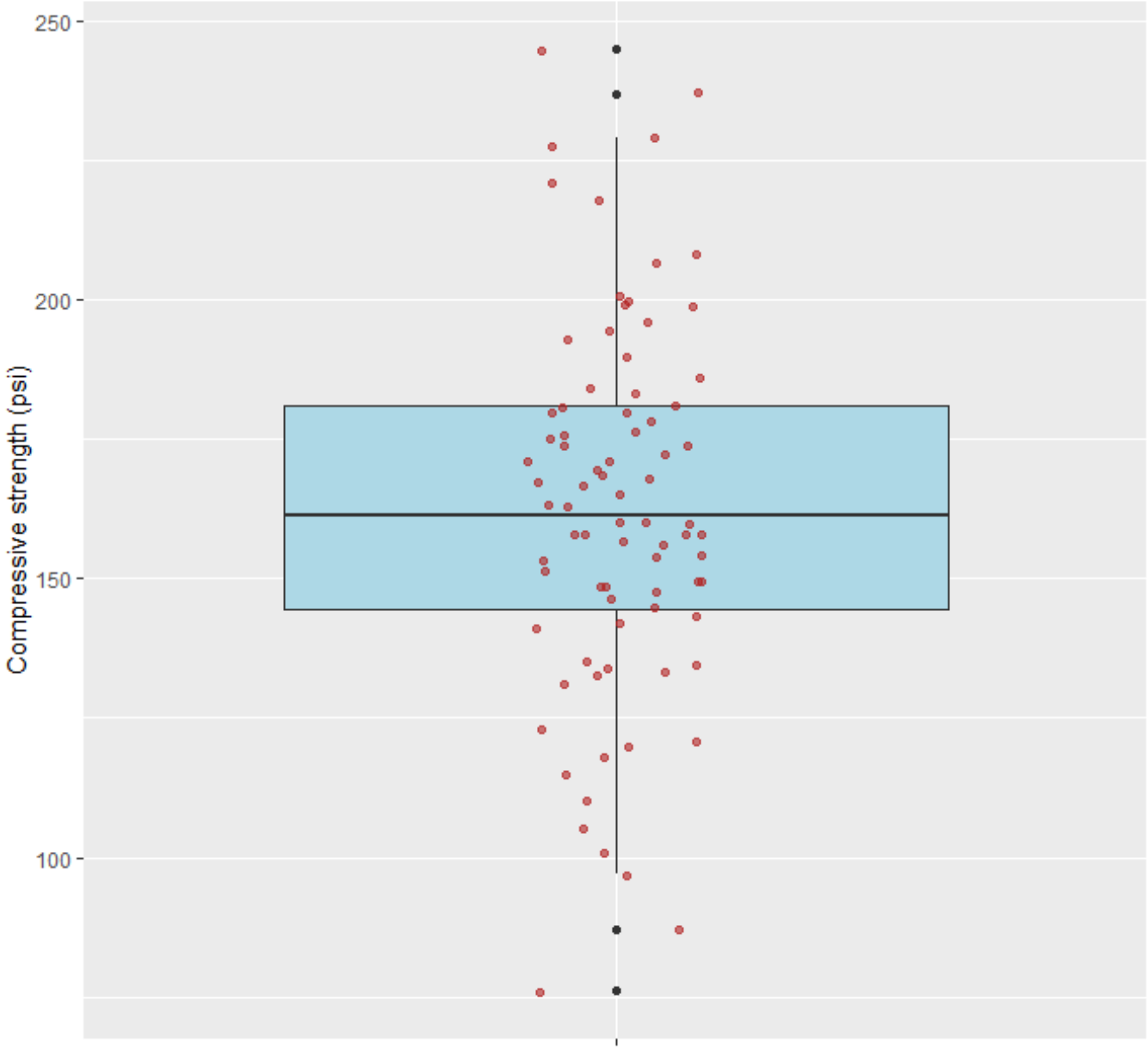


Figure 2-13: Compressive Strength Boxplot (ggplot2)



- **Fig. 2-14** demonstrates how comparative box plots effectively highlight differences in quality and variability across multiple data sets.

  - **Comparative Power**: As shown in **Fig. 2-14**, box plots provide a high-impact visual for contrasting performance across different groups or locations.

  - **Variability Analysis**: The display in **Fig. 2-14** reveals that Plant 2 suffers from excessive variability compared to the others.

- **Performance Gaps**: Analysis of the plots indicates that both Plant 2 and Plant 3 are underperforming and need to improve their quality index levels.

Figure 2-14: Tensile Strength Comparison (Base R)

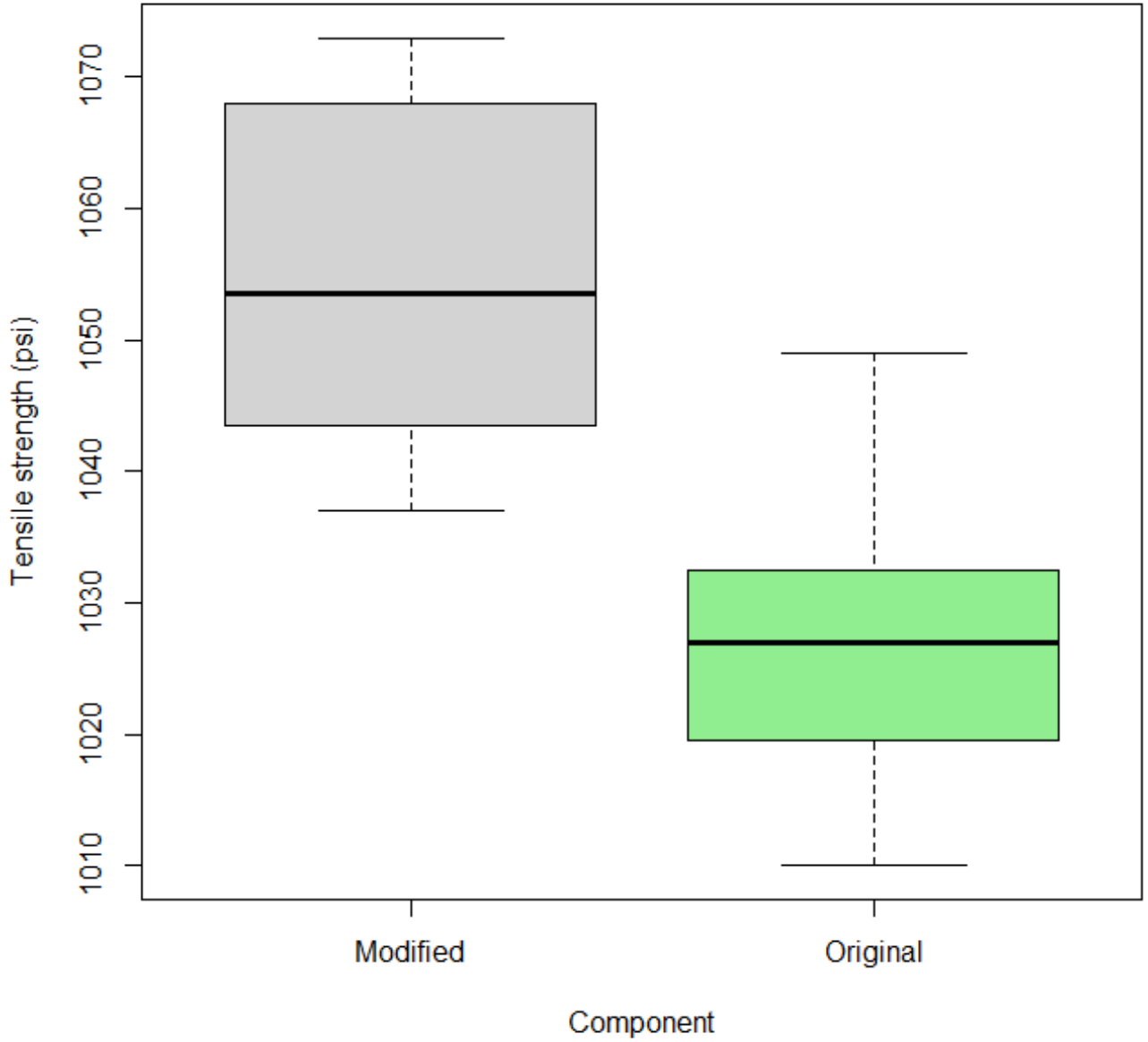
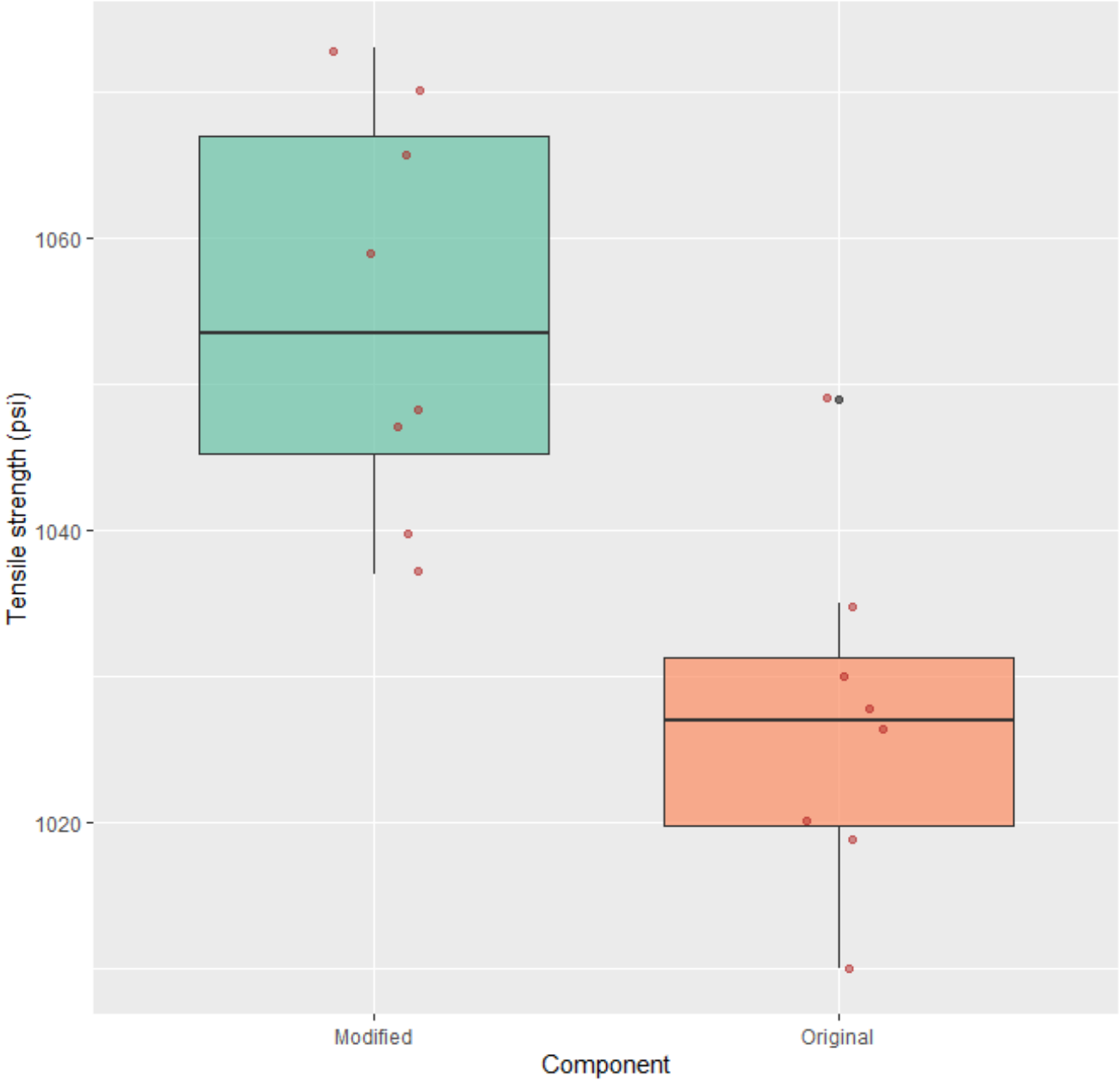



Figure 2-14: Tensile Strength Comparison (ggplot2)